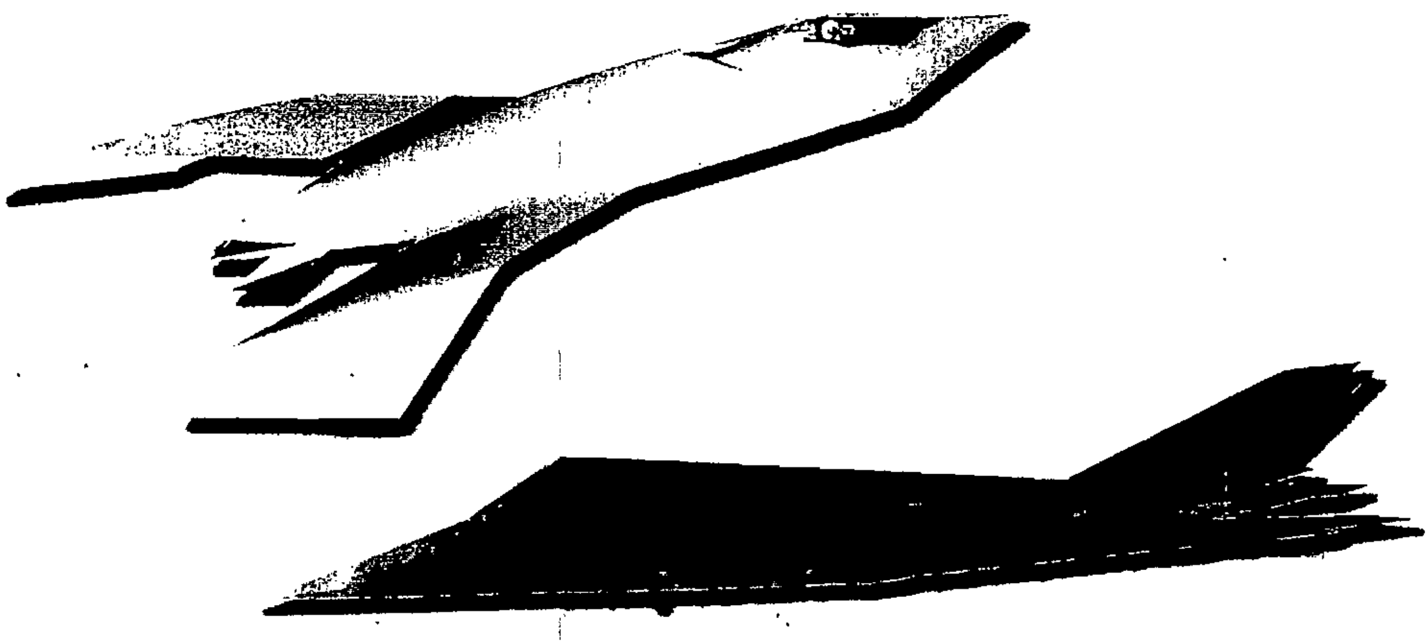
FLIGHT TEST HISTORICAL FOUNDATION

Gathering of Eagles 2004

Out of the Black . . .
. . . Into the Blue

Paper Designs
to
Flying Prototypes



October 1, 2004

The 2004 Eagles

[redacted]

[redacted] grew up in La Crosse Wisc., and graduated from the U.S. Air Force Academy in 1978 and completed pilot training at Williams AFB, Ariz.

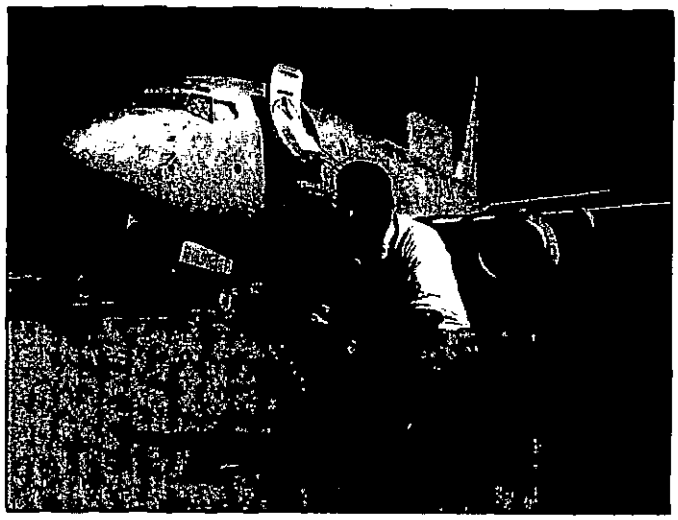
Benjamin stayed on at Williams as a T-38 instructor pilot, and then flew operationally in the F-106 at K.I. Sawyer AFB, Mich., and then the F-16 at Homestead AFB, Fla.

He attended EPNER, the French Test Pilot School, in Istres, France, and then returned to Edwards AFB, Calif., to first instruct in the test management branch at the USAF Test Pilot School and then managed the flying qualities branch. Benjamin then moved to the F-16 Combined Test Force, testing avionics upgrades on all F-16 models and kicking off testing on Digital Flight Control System improvements for the F-16 Block 40 and 50.

After Air Command and Staff College at Maxwell AFB, Alabama, and Defense Systems Management College at Fort Belvoir, Virginia, Benjamin served a tour of duty at the F-16 Systems Program Office at Wright-Patterson AFB, Ohio, as first chief of multinational test for 21 different foreign military sales test programs and then served as deputy director of the F-16 flight test division.

Benjamin then moved to the black world where he flew on and commanded a variety of classified programs. He was the second of only three pilots to fly the Boeing Bird of Prey and flew 21 of its 38 flights, including envelope expansion, mission utility and tactical applications.

[redacted]etired from the Air Force in 2000 and became an engineering test pilot with Boeing Commercial Airplanes in Seattle,



Wash. Since arriving at Boeing, he has flown on the 737-900 and 777-300ER development and certification programs, as well as the 757 F-22 Flying Test Bed. He was project pilot for development and certification testing on the 737 of a new digital auto flight system, a GPS landing system, new elevator tab hinges and the Rudder System Enhancement Program. He's most recently been flying on Project Wedgetail, an Airborne Early Warning and Control version of the 737-700. He is currently the 737 senior project pilot. He has more than 6,750 hours of flying time in 89 aircraft and is current on all Boeing models, as well as the T-33 and the T-38.

He's been a member of the Society of Experimental Test Pilots since 1996 and an Associate Fellow since 2001.

[redacted]

[redacted] began his aviation career at the age of 14 when he used paper route earnings to pay for his first flight in an Aeronca 7AC. That flight was enough to kick off the start of a forty plus year journey that is continuing today.

[redacted]received a B.S. degree in Aircraft Maintenance Engineering from

Parks College of St, Louis University in 1963. He chose Parks College in part because it offered Air Force ROTC, a big consideration due to the fact that the draft was active and he knew the route to a professional pilot career could be assured with Air Force experience.

He was then was commissioned in the U. S. Air Force and served six years in the USAF before joining McDonnell Douglas as a Flight Safety Engineer. He later earned a MS in Engineering.Management from the University of Missouri-St. Louis.

In May 1976, Haug became a production test pilot with McDonnell, flying the F-4 Phantom and the F-15 Eagle. He was the primary pilot in the development of an Instrument Landing System for the F-4 series aircraft. He was active in experimental testing of the F-15 including overload warning system, radar development, and several programs in the Phantom Works.

He also participated in the Full Scale Development program testing of the F-18 Hornet, including increased roll performance, G-limiter system, and flutter with external stores.

A great deal of his flight hours were accumulated in the Missouri Air National Guard, where he flew the F-100 and the F-4 until 1989, when his 26 years of military service came to an end as the 110th Tactical Fighter Squadron Operations Officer.

He became Chief Test Pilot in 1986 and was responsible for Boeing's Military Aircraft and Missile Systems flight operations of the AV-8, F-18, F-15, T-45 and King Air Test Bed aircraft. Haug was the primary and first flight pilot of the Bird of Prey technology demonstrator aircraft, until retirement this year.

[redacted] has been a member of SETP since 1985 and is active in general aviation as a CFI; A&P mechanic, aviation consultant, aircraft owner, and regularly flies a Baron for travel and a 450 Stearman for fun.

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[redacted]

[redacted] was born and spent his formative years in in New York. He earned a B.S. degree in aeronautical engineering at Parks College of Aeronautical Technology of St. Louis University in 1952. Later, he graduated in the class of 31 from the [redacted] Pilot School.

[redacted] spent five years in the Kansas Air National Guard concurrently with his Beech and Boeing employment.

His flying career began with the USAF in 1952 where he graduated primary flight training class 53D at Marana, Ariz., flying T-6's and transitioned to T-33s at James Conley Air Base, Waco, Texas. In all, Thomas spent four years as an All Weather Fighter Pilot.

He attended combat training at Tindel AFB in Florida, flying F-86Ds and was later assigned to an F-86D squadron at McCord AFB, Wash.

[redacted] began his test pilot career with Beech Aircraft as production acceptance and demonstration pilot in Beech Bonanzas and Travelairs.

In 1958 he joined the Boeing Co. in Wichita, KS, as an experimental test pilot working on ERB-47s and B-52s. Thomas also did production flight testing and participated in the design proposal of the TFX.

In 1963 [redacted] began his career with Northrop as an engineering test pilot flying a vast variety of aircraft. He was supervisor of engineering test pilots at Northrop's Norair facility at Edwards AFB, Calif. and has done extensive testing with the F-5 program.

[redacted] worked on the development of Tacit Blue and was the first pilot the aircraft.

[redacted] left Northrop in 1970 and opened his own aerospace consultancy business shortly thereafter.

Thomas has flown over 35 types of aircraft in his career, including the F-5A/B, B-52E/F/G.H and the X-21.

Now retired, Thomas makes his home in Lancaster, Calif. and remains an active member of the Society of Experimental Test Pilots, the American Institute of Aeronautics and Astronauts and the Quiet Birdmen.

[redacted]

[redacted] has been the lead pilot on seven classified aircraft to date. His has been a career filled with few honors and many private successes.

In his career, many of his flights have been conducted in one-of-a-kind technology demonstrator aircraft. He has flown first flights with modified landing gear, first flight with internal bays installed, first flight where bay doors were opened, and initial separation tests. He holds the altitude record in this aircraft.

He is one of the five pilots who flew Northrop Grumman's Tacit Blue. Since [redacted] has made his career in the cockpit of so many classified aircraft, there is not much that we can say about him, on the record. In letters of recommendation, it is noted by his superior officers that his work has been outstanding and will probably never be recognized by the general public.

[redacted] is a graduate from the University of Idaho and is a member of the Society of Experimental Test Pilots.

He is currently working at Edwards AFB on a classified program.

Flight Test Historical Foundation

Welcome to the 2004 Gathering of Eagles

: Directors and Trustees of the Flight Test Historical Foundation e great pleasure in welcoming you to the 12th annual Gathering of gles. Thank you for joining us tonight to celebrate the achievements he flight testers associated with three technology demonstrator totypes; code named Have Blue, Tacit Blue and Bird of Prey.

st year we celebrated the 100th anniversary of the first controlled, wered, heavier than air flight. The Wright's careful, measured, entific method approach to prototyping the Wright Flyer airplane ulted in the outstanding achievements in proving and developing ir concept of powered flight. The Wrights set the engineering ndard for experimental flight test that is still used today.

l three of the prototype aircraft programs we are recognizing present an interesting balance of performance, stability and control d low observable "stealthiness." Often, the "mix" of stealth and rformance and flying qualities (P&FQ) requirements presents a mendous engineering challenge in reaching both the low observable als and acceptable P&FQ. Despite the use of very powerful alytical design tools the only way to truly prove an advanced ncept is to build and test a prototype 'first of a kind' aircraft. The ave Blue prototype proved that the full scale development 'faceted' -117 Stealth Fighter was a sound and practical possibility, and the cit Blue prototype proved that the 'smooth' stealth concepts used on e B-2 could be implemented. All three of these outstanding ototype technology demonstrator aircraft and their flight test crews ve dramatically advanced the state of the art of aviation!

he Flight Test Historical Foundation has had considerable success in fundraising to benefit the Air Force Flight Test Center AFFTC) Museum. We funded the construction of the 12,000 square foot AFFTC Museum, and the recovery and restoration of any priceless and historical flight test aircraft. In addition, we supported improvements to and restoration of many exhibits at both e BlackBird Airpark and AFFTC Museum. This year, the FTHF funded the construction of a brand new gift shop and office cility at the BlackBird Airpark. We owe all of you heartfelt thanks for your continued support and contributions that have helped nd the facilities and many outstanding exhibits and interpretive displays at the museum.

alifornia State Senator 'Pete' Knight conceived the FTHF idea while he was serving at Edwards AFB as a Colonel in the US Air orce, specifically, as Vice Commander of the AFFTC. We would like for you to keep our recently departed friend and FTHF rustee Pete Knight in your thoughts tonight.

Ve would be remiss if we did not recognize the hardworking FTHF GOE Committee members, FTHF Board members and Trustees, AFFTC staff members and most especially our Corporate Sponsors. Finally, special recognition is due AFFTC Museum Curator Doug Nelson and his wife Ilah for their outstanding spirit of cooperation and unceasing work at the Museum.

Thanks again for your outstanding support and we hope you enjoy the 2004 Gathering of Eagles.

Warmest Regards,

[redacted]

Black World Trailblazers
Out of the Black Into the Blue

[redacted]
Director, Flight Test Historical Foundation

The 1970's led to many changes in American culture. In addition to disco, leisure suits, and platform shoes, a more lasting revolution in aerospace technology appeared on the scene; but it was initially only known to a select few. The revolution was what is known commonly as " stealth technology", and it resulted from collaboration between the American defense department and the aerospace industry. It has affected the way wars were fought and will be fought into the future. Much of the pioneering work in this field is only becoming declassified today, and some still remains secret due to its importance. We salute all the people who made this happen by choosing three programs that laid the foundation for a technology that has such a major influence on aerospace designs as we enter the second century of flight. For years they could not tell even their families what they were doing. This is part of their stories.

Stealth: The Background

Aerial combat experience in Vietnam and in the Middle East signaled to the Defense Department and the Air Force that things were changing. The 1950's vintage Russian SA-2 surface to air missile system (SAM), although having a success rate of less than 5 %, had significantly inhibited aerial operations over North Vietnam. The loss of 40 Israeli combat aircraft to more modern SA-6 SAM and ZSU 23-4 radar guided antiaircraft guns in the first 2 days of the 1973 October War showed the problem of operating against an integrated air defense system would get only worse. As a result of these concerns, the Defense Advanced Research Agency (DARPA) in 1974 requested studies from aircraft manufacturers. Since the air defense systems generally relied on radar or infrared heat detecting sensors to acquire and track targets, there were two questions to be answered: a) what level of reduction in the radar and infra red signatures of an attacking aircraft would be necessary to achieve a significant tactical advantage over the air defenses; and b) could an airplane be built to achieve the required low signature levels? The concept would lead to what became known as low-observability (LO) designs, popularly known as " stealth" designs.

Northrop and McDonnell Douglas were given study contracts to answer the questions. Lockheed was initially not included since they had not built tactical aircraft in over 10 years, but they were also invited after they revealed they had been investigating stealth features for the CIA-initiated U-2 and Blackbird reconnaissance planes since the late 1950s.

The studies revealed that a large reduction in signatures over contemporary designs would be necessary to make a significant difference. The longest range sensor was radar; and the radar cross section (RCS), as the measurement was commonly known, of an attacker would have to be reduced by orders of magnitude to have an appreciable effect. Ever since radar was invented in World War 2 methods of measuring the RCS of an airplane had been developed, but few attempts had been made to reduce it in the aircraft design process. A reduction of over a hundred fold seemed unachievable. Nevertheless, Northrop and Lockheed believed there was a possibility. In 1975 the 2 companies were funded to produce competing designs for an Experimental Survivable Testbed (XST) stealth demonstrator. Initially full scale models of the designs were produced, mounted on poles, and tested on outdoor RCS measuring ranges . The Lockheed design produced the better results, so they

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[redacted]ails from Albany, New York, and received his Bachelor's of Aeronautical Engineering from Rensselaer Polytechnic Institute, Troy, New York in 1955. He later earned a MBA from Pepperdine College in 1970.

[redacted] began his professional career at the Lockheed Aircraft Company's Flight Test Division, where he was assigned to work with propulsion and fuel lab technology.

Just a year later, he joined the USAF as an Aviation Cadet and took his flight training at Malden Air Base, Missouri.

In 1957, he to Lockheed to join Kelly Johnson's Skunk Works Flight Test organization, and was assigned to the U-2 test program at Edwards AFB, where he worked until 1960 when he was reassigned to Burbank, to begin test & development planning for the A-12 aircraft, (forerunner of the SR-71).

In 1962, Beswick found himself transferred to begin work on the A-12 flight test program. He also flew on many flights as an engineering observer and launch control officer on the MD-21 program.

In October of 1966 he returned to Burbank once again and joined Lockheed Flight Test's Advanced Programs Dept. to begin test planning for the Navy's VSX test program, (which later became the S-3A).

After completion of a one-year Inter-branch Rotational Management Training Program, [redacted] found himself promoted to Department. Manager, in Lockheed's L-1011 Flight Test organization in 1970 where he worked as the Assistant Director of L-1011 Quality Assurance, and later was promoted to Director of L-1011 Quality Assurance.

In. 1975 [redacted] was assigned as Manager of Flight Test and Rye Canyon's Administrative Organization, Valencia, California. Followed closely by a stint as the Director of Engineering at the Lockheed Aircraft Service Co. in Ontario, California.

[redacted] returned to Lockheed's Skunk Works organization in 1976 to begin test planning for the "Have Blue" Stealth Demonstration Program.

In. 1978 he was promoted to Director of Flight Test at Lockheed and became responsible for the U-2, SR-71, F-117A, F-22 and several other classified programs.

In March 1991, [redacted]k retired after 36 years with Lockheed Martin where he left a legacy that began at Kelly Johnson's side, and continues with the F/A-22 as it makes its mark as a premier fighter plane of the future.

[redacted] *USAF* [redacted]esting of top secret aircraft put him on the ground floor of stealth technology, flying Have Blue, the prototype for the F-117A Stealth Fighter, and Tacit Blue, which demonstrated radically different stealth technologies from Have Blue.

After four years as a fighter pilot, [redacted]ttended the USAF Aerospace Research Pilot School prior to testing weapons in the F-100, F-101 and F-4 aircraft. He flew the F-100 and F-4 aircraft in the Vietnam War before returning to Edwards AFB as an instructor at the USAF Test Pilot School and later as an F-15 test pilot and Director of the F-15 Joint Test Force. He began classified work in 1976, where he flight tested Have Blue and Tacit Blue until 1982.

After Air Force retirement, [redacted]n joined Rockwell and flew throughout the B-1B test program. He flew the first flight of the X-31 Post Stall Aircraft in 1990 and flew the X-31 through its early testing. He retired from Rockwell as Chief Test Pilot and Director of Flight Test in 1993.

[redacted]was awarded the Kincheloe Award in 1989 for test flying Have Blue and in 1996 for Tacit Blue after these programs were declassified. He has also received the Legion of Merit, two Distinguished Flying Crosses, nine Air Medals and Aviation Week & Space Technology's Aerospace Laurels. Dyson is an Engineering Fellow of the University of Alabama, a Distinguished Alumnus of the USAF Test Pilot School and was named to the Aerospace Walk of Honor in 1997.

were awarded the contract to produce 2 actual aircraft to validate the pole model results. In addition, as it was realized it might be possible to actually produce a combat airplane, the entire idea of stealth aircraft moved into the " black world" of covert programs with information limited to those with a demonstrated " need to know. The XST flyable demonstrator was to be code-named Have Blue.

Have Blue; Lockheed's Hopeless Diamond Takes Wing

The designs produced by the competitors were similar in many features, but different in the details. Studies had shown that the important feature was to reduce the signature when viewed from the front or from the rear, with the front being most important for an aircraft penetrating the hostile air defense system. Northrop had used prior experience to develop ideas, then test them with small models on indoor RCS ranges. Lockheed, on the other hand, had discovered a paper written by a Russian mathematician (and translated by the USAF Foreign Technology Division and republished in the United States) that developed equations to approximately predict the RCS of simple shapes. The same computers that solved the equations could be used to optimize (actually minimize) the RCS of a collection of simple shapes that constituted the aircraft design. The optimization process, coupled with the designers refinements, produced a design with unparalleled low signatures from certain look angles— but a design that looked at first glance like it would never fly, figuratively and literally. The computer design didn't have any wings. The Lockheed aeronautical engineers dubbed the configuration the " Hopeless Diamond"- they didn't think there was any way to turn it into a practical airplane. But with modifications, maybe it was possible.

Two Have Blue airplanes were built. One concentrated mainly on the flying qualities of the design; the second was intend to confirm the extraordinarily low signatures of the pole model with an actual airplane in flight. The flying qualities were especially important, because the RCS -driven design had produced an airplane that seemed to violate all laws of aerodynamics. It had a " faceted" look (the diamond planform wasn't the only reason for the " Hopeless Diamond" nickname) consisting of numerous flat planes. Although intended to reflect reflected radar pulses back in different directions from the location of the transmitting radar, the faceting also meant the airplanes did not have any smooth curves that aerodynamicists strive for to allow good performance. Although a twin-engined jet aircraft, the airplane at first glance did not seem to have the inlets or exhausts that are so obvious on most designs. Fine-meshed screens to conceal the rotating engine compressor blades from hostile radar pulses covered the inlets. The exhaust, rather than the traditional round nozzle, was a less obvious" letterbox slot" rectangular orifice, intended to hide the rotating engine turbine from radars behind the aircraft, as well as reduce the heat signature of the exhaust gas stream. The highly swept wing necessary to retain the diamond planform meant the aircraft would have a high stall speed, with a high landing speed, making it a " hot ship" in the landing pattern. The inward-bent vertical tails meant that if the airplane did slow to the point it achieved high angles of attack, the tails would

probably be blanked by the disturbed air from the wing wake, leading to poor directional control. All in all, the design had a number of compromises thanks to low-RCS design; the question was would it all work?

The Have Blue aircraft were intended strictly to prove the concept of stealth. Too small (only 38 feet long) to carry a weapon payload, they saved costs by " borrowing" parts from other airplanes. The landing gear was from an A-10, the ejection seat from an F-5, and the flight controls computer (FCC) from an F-16. The latter was significant; in 1977 the F-16 fly-by-wire concept was still new. The F-16 FCC was fitted to an airplane that looked completely unlike the " parent" aircraft, not even having a separate horizontal tail. Reprogramming was necessary to account for the differences. Nevertheless, a small team of engineers and technicians built the two vehicles at Burbank in a matter of months. A machinists strike during the construction phase was an additional challenge; it meant that much of the hands-on work was actually performed by the managers and supervisors such as [redacted] the latter was actually head of the flight test effort. . The need for absolute secrecy was also a challenge; it required nighttime operations and masking of the aircraft whenever it was outside its construction building. A unique camouflage paint scheme was initially used, designed to deceive the eye so the faceted aspect of the design was not obvious. Once built, the small vehicles were loaded into a C-5 Galaxy cargo plane and flown out (late at night, which is normally verboten at the Burbank airport) to the desert flight test location.

The first Have Blue was flown by veteran Lockheed test pilot, Bill Park on Dec. 1, 1977. The takeoff run was long, and the rate of climb was low. The non-afterburning J-58 engines did not provide an excess of thrust; nevertheless, the airplane's limited performance envelope was adequate to confirm the basic design was flyable. The low RCS of a stealth aircraft meant the radar reflections produced by a normal pitot-static system with its airspeed measuring pitot boom had to be reduced. The success of the new pitot-static system design had to be verified on ship one, which had a conventional pitot boom, installed for the first flights as a back up. The flight envelope exploration continued through May of 1978, with Air Force test pilot [redacted] sharing the piloting duties. The program stopped abruptly on 4 May 1978, when a flight control system-induced pitchdown occurred as [redacted] was performing a low-speed landing. The airplane slammed onto the runway. Park performed a touch and go and climbed out to evaluate possible damage. The impact had damaged the landing gear resulting in the right main landing gear being jammed up, meaning a safe landing was now impossible. After numerous attempts to extend the jammed gear, running low on fuel to the point that one of the engines flamed out, [redacted] ejected and Have Blue 1 crashed into the desert. Park himself was injured during the ejection to the extent that he never flew as a company test pilot again.

Have Blue 2 first flew in July 1978. It has all the features designed to reduce the signature. It demonstrated that an airplane in flight could replicate the pole model results, despite being stretched and deformed by the lift and drag forces to which it was exposed. Practical flight test experience also demonstrated the difficulties in maintaining the many features besides the shape that contributed to the extremely low RCS. Sheets of Radar Absorbent Material (RAM), similar to linoleum, covered much of the aluminum airplane replacing those when maintenance panels were opened required special attention. Filling in the tiny gaps and seams around the access panels and even the canopy also was time consuming. Even a loose screw could show up on radar, requiring reflying the test. The "devil was in the details" for maintaining the low RCS, but the flight test technicians showed it was possible, if laborious.

But the results were worth the effort. Flown against many radars, including Soviet-style systems, the overall design was shown to produce a significant reduction in detection range. This in turn showed that a tactical advantage could be achieved by exploiting the gaps opened up in an air defense system designed to detect non-stealth aircraft. The Have Blue was allegedly even flown in close proximity to an E-3 AWACS radar plane whose screens barely showed its presence.

Based upon the results, the Senior Trend program was initiated to produce a weapon system based upon the Have Blue design. Considerably larger and with an internal weapons bay designed to carry two laser-guided 2000 lb. bombs, the resulting aircraft became the F-117. The success of the concept was proven in the Desert Storm opening night attack, when the F-117s were assigned to attack the most heavily defended targets, including downtown Baghdad. Pilots admitted afterward as they watched the AAA guns firing and the missiles launched over the city, that they had misgivings over whether " this stealth really worked". The answer was: it did. The guns and missiles were all launched blindly. Throughout the entire air campaign, no F-117s were ever hit despite being in the thick of the battle night after night. Stealth was now a combat-proven concept.

Have Blue 2 was not around to bask in the success. On a test mission one year after its first flight, a duct containing the hot jet exhaust within the fuselage failed, releasing the hot gasses into the aircraft and initially destroying one of the two hydraulic systems. Pilot [redacted] attempted to return to base and land, but soon the second hydraulic system went the way of the first, leaving him without flight controls in his fly-by-wire airplane. He ejected safely, and the second aircraft followed the first into the desert sands. But the test program was almost over anyway, and the Have Blue mission had been accomplished. Only a few photos, short films and memories remain of a trailblazer in every sense of the word.

TACIT BLUE:
"Got a Whale of a Tale to Tell Ya' Lads——"

The above line, sung by Kirk Douglas in the 1950's Disney movie production of 20,000 Leagues Under the Sea, could not have been more appropriate-especially considering the Disney connection that will be described later. For well over a decade "the whale" served as an unofficial symbol to commemorate another chapter in the history of the revolutionary concept of low stealth. The official name of the program was Tacit Blue; and like Have Blue, it was intended to demonstrate concepts that were revolutionary as well as untried when the idea was conceived.

Although Northrop had lost to Lockheed in the competition for the XST, many valuable lessons had been learned. DARPA had been impressed by their entry for the XST, and awarded them a contract for a study for a project known as BSAX, for Battlefield Surveillance Assault- Experimental. The concept was to evaluate the feasibility of building a stealthy radar-equipped aircraft to loiter in the projected World War III high threat battlefield of Central Europe. It was to detect and track Soviet tank columns for prosecution by missiles launched by allied ground and air forces. This proposed concept of operations would require the airframe to be stealthy, but not just from the front, as the Have Blue / F-117 design, but from all directions, since the aircraft would be in a racetrack-shaped orbit near the battle-lines so it could use its high-powered SLAR (Sideways Looking Airborne Radar). The challenge of stealthy radar was another problem, since ordinary radar antennas can have a huge RCS, as well as radar transmissions being detectable by sensors on the ground. Northrop teamed with Hughes (now Raytheon) on development of the radar and solving the problems associated with making its transmissions low observable.

The airplane configuration was largely driven by the need to mount the large flat radar antenna looking out the side of the fuselage. This produced a boxy fuselage, with initially flat sides. Model tests showed this produced a large RCS. Sloping the sides reduced the RCS considerably, similar to the faceted flat plates of the Have Blue. But the real problem was how to reduce the RCS when the threat radars were illuminating the vehicle in 3 dimensions, rather than from only one direction at a time as the Have Blue approach was optimized for. The flat plates of the Have Blue would not do; curved Gaussian surfaces were another approach to redistributing the reflected radar pulses away from the radar transmitter. The computers of the late 1970s were not yet able to optimize a design in 3 dimensions, so more experimentation was required to evaluate and modify proposed designs. The RCS engineers were in the habit of using modeling clay to visualize proposed configurations in 3 dimensions to help them think out their ideas. One engineer, Fred Oshira, who was working on the forward fuselage design, took his children to Disneyland one weekend, and while waiting for them to finish a ride, he was struck by a possible solution, which he quickly modeled using the clay he happened to still be carrying with him. Allegedly the ride was the Mad Hatter's Teacup ride. For those who remember Disneyland in the 70's, recall what was across the street from that ride,.the Storybook Land Ride, with the huge Monstro the Whale tunnel through which the boats traveled. The whale's head is a complex curved shape, equivalent to a 3-dimensional surface that would produce radar reflections in all 3 dimensions. Subsequent model tests of Oshira's lump of clay design showed this shape was the beginning of minimizing the RCS from all directions ; furthermore, it allowed use of curved surfaces which the aerodynamicists desperately wanted if stealth airplanes were ever to also be efficient airplanes. If ever there was an engineering " Eureka" moment in modern times, this may have been it.

The single Tacit Blue airplane was built in Hawthorne, California. It was rolled out in 1982 and immediately drew as many stares as the Tacit Blue had when people first saw it. At 55 feet long and weighing 30,000 pounds it was not exactly petite; and the final